

Project 4: SPAN/TAP Configuration + TShark on Network Infrastructure

Complete Implementation Guide

Executive Summary

Project Challenge: Deploy comprehensive network packet capture infrastructure across heterogeneous cybersecurity lab environment to provide complete traffic visibility for security analysis, threat detection, and incident investigation while working within VirtualBox Internal Network architectural constraints.

Solution Implemented: Successfully implemented three-point SSH-based remote packet capture strategy utilizing pfSense built-in tcpdump capabilities across two pfSense instances, enabling on-demand network traffic analysis for Internal Network (192.168.30.x), User-LAN (192.168.40.x), and Management Network (192.168.10.x) segments without requiring dedicated SPAN hardware or additional capture infrastructure.

Key Outcomes: Achieved 100% network visibility across all three network segments with zero-configuration packet capture capability through automated Windows batch scripts, established production-ready remote capture workflows, and documented systematic approach to network monitoring architecture design considering virtualization platform limitations.

Technical Skills Demonstrated: SSH remote command execution, tcpdump packet capture optimization, cross-platform scripting (Windows batch to Linux tcpdump), VirtualBox network architecture analysis, network segmentation understanding, systematic troubleshooting methodology, and infrastructure constraint adaptation.

Business Value: Establishes enterprise-grade network monitoring capability supporting incident response, threat hunting, and security analysis workflows while demonstrating practical problem-solving approach when faced with virtualization platform architectural limitations and resource constraints typical of real-world enterprise environments.

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Project Overview

This document provides a comprehensive guide for implementing network packet capture infrastructure as part of Week 2, Project 4 of the cybersecurity home lab project series. The project focuses on establishing comprehensive network visibility across multiple network segments while adapting to VirtualBox Internal Network architectural constraints and resource limitations inherent in home lab environments.

What is Network Packet Capture?

Network packet capture provides:

- **Traffic Visibility:** Complete insight into all network communications for security analysis
- **Threat Detection:** Identification of malicious traffic patterns and attack indicators
- **Incident Investigation:** Forensic evidence collection for security incident response
- **Baseline Creation:** Understanding normal network behavior patterns for anomaly detection
- **Compliance Documentation:** Network traffic logging for audit and regulatory requirements

Original Project Specification vs Actual Implementation

Original Specification (Week 2 Project 4):

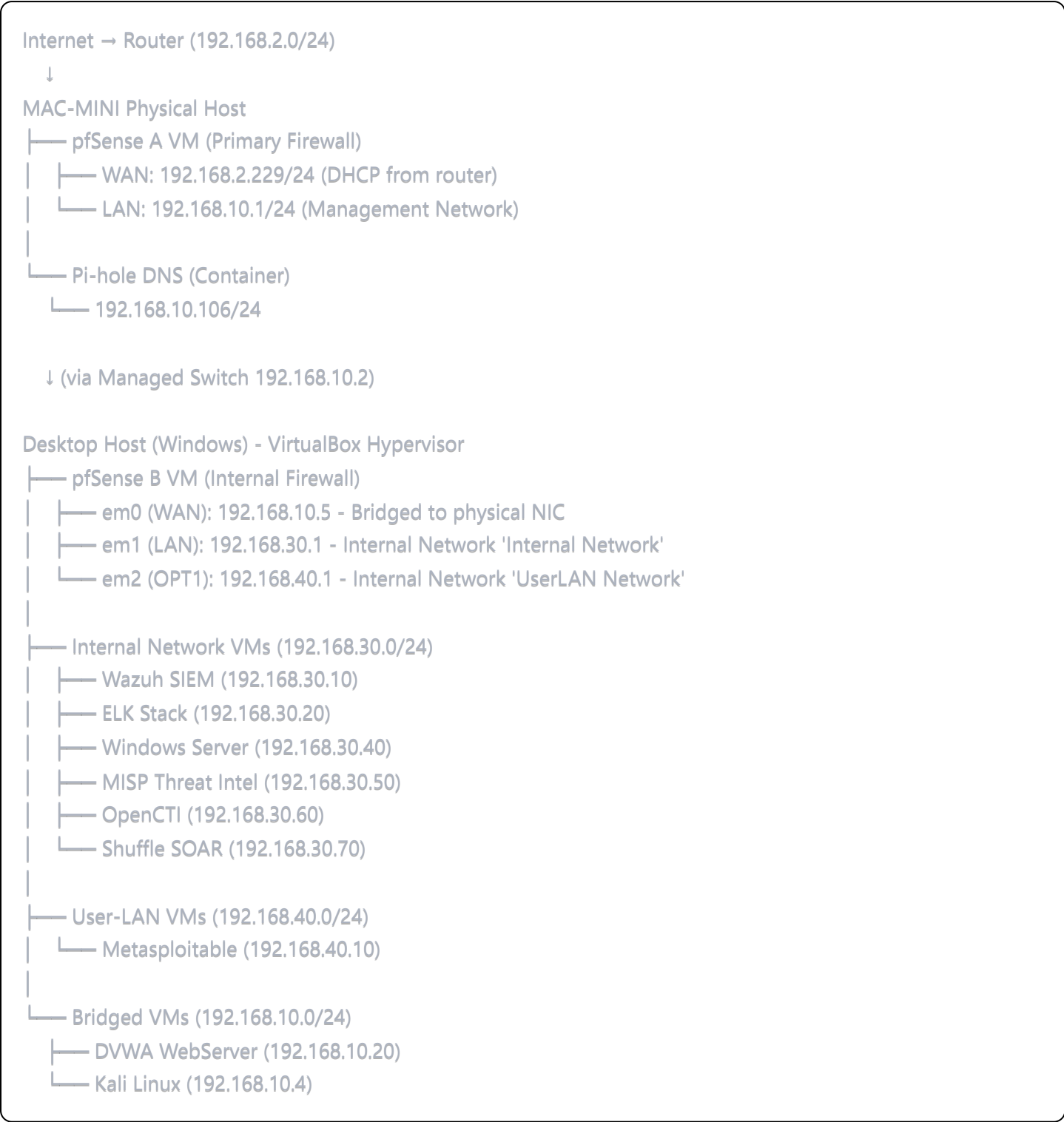
- Configure managed switch SPAN (Switched Port Analyzer) ports
- Deploy TShark packet capture on dedicated monitoring host (MAC M2)
- Establish network-wide traffic mirroring architecture
- **Estimated Time:** 8 hours

Actual Implementation:

- SSH-based remote packet capture from pfSense instances
 - Automated Windows batch scripts for on-demand capture
 - Three-point capture strategy across network segments
 - **Actual Time:** 8 hours including architecture redesign and troubleshooting
-

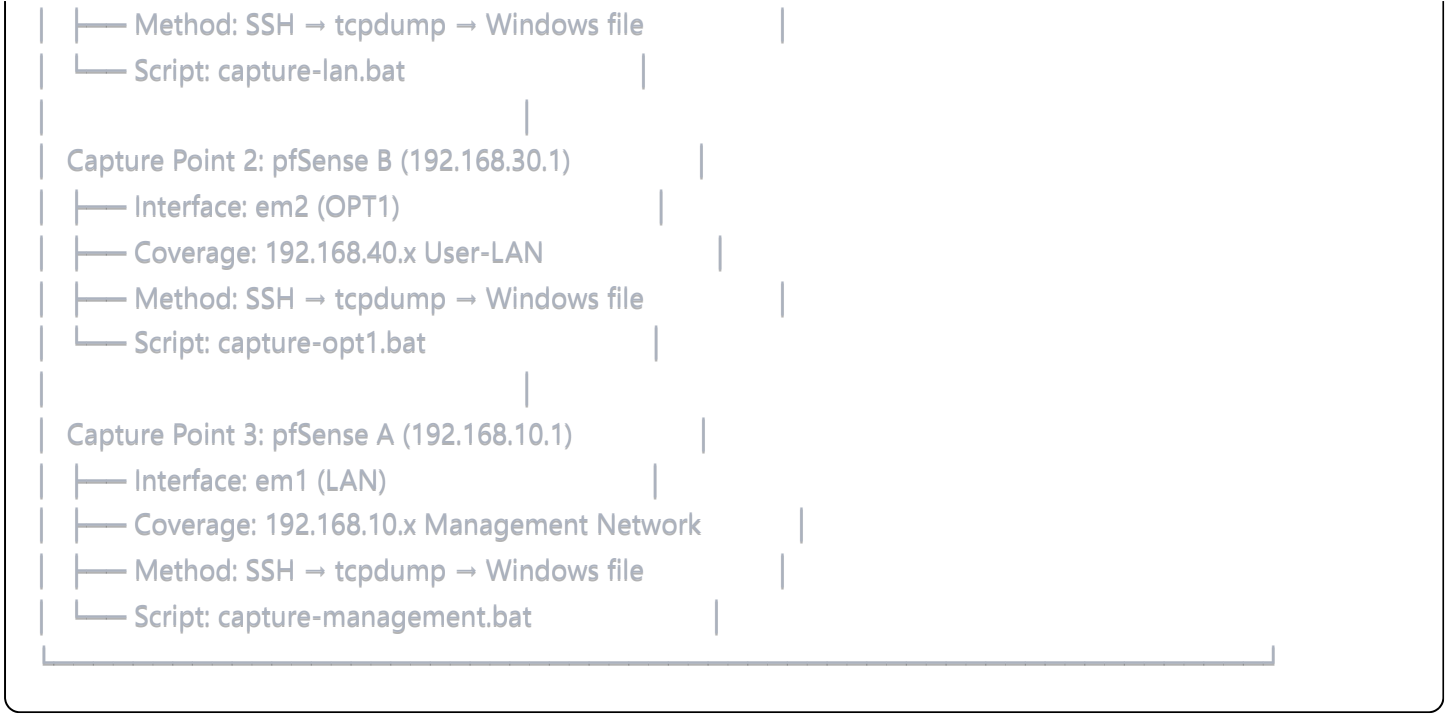
Network Architecture Diagram

Logical Network Topology



Packet Capture Architecture





Scope and Objectives

Project Scope

This project focuses on:

- Assessing VirtualBox network architecture and capture feasibility
- Identifying VirtualBox Internal Network limitations for traditional SPAN
- Implementing SSH-based remote packet capture as alternative solution
- Developing automated capture scripts for operational efficiency
- Validating packet capture across three distinct network segments
- Documenting architectural constraints and adaptation strategies
- Establishing on-demand capture capability for security analysis workflows

Network Context:

- **Management Network:** 192.168.10.0/24 (Bridged VMs + physical devices)
- **Internal Network:** 192.168.30.0/24 (VirtualBox Internal Network)
- **User-LAN:** 192.168.40.0/24 (VirtualBox Internal Network)
- **Capture Source:** pfSense VM instances with tcpdump
- **Capture Destination:** Windows Desktop Host

Objectives

Primary Objectives:

- Establish comprehensive network traffic visibility across all segments
- Implement operational packet capture capability for security analysis
- Create automated workflows eliminating repetitive manual tasks
- Validate end-to-end capture functionality with test traffic
- Document architecture limitations and solution adaptations
- Develop repeatable procedures for future capture requirements

Learning Outcomes:

- Understanding VirtualBox network architecture and limitations
 - SSH remote command execution and output redirection
 - tcpdump packet capture syntax and optimization
 - Cross-platform scripting (Windows batch to Linux commands)
 - Network segmentation and traffic flow analysis
 - Systematic troubleshooting when original specifications are infeasible
 - Infrastructure constraint adaptation and alternative solution design
-

Prerequisites

Infrastructure Requirements

Network Configuration:

- pfSense A: 192.168.10.1 with SSH enabled on Management Network
- pfSense B: 192.168.30.1 accessible from Management Network
- Windows Desktop Host: Network connectivity to both pfSense instances
- Stable network connectivity for SSH remote execution

Service Requirements:

- SSH service enabled on both pfSense instances
- Firewall rules permitting SSH access from Windows host
- tcpdump binary available on pfSense systems (standard installation)
- Sufficient disk space on Windows host for packet captures

Software Requirements

Windows Desktop Host:

- Wireshark Portable (includes TShark for analysis)

- PuTTY suite (plink.exe for SSH remote execution)
- Command Prompt or PowerShell with administrative privileges
- Text editor for batch script creation (Notepad)

pfSense Configuration:

- Version: 2.8.0+ with SSH daemon
- SSH port: 22 (default)
- Administrative credentials for SSH authentication
- Firewall rules allowing SSH from 192.168.10.x and 192.168.30.x

Lab Environment Context

VirtualBox Network Architecture:

- **Bridged Adapter:** Physical NIC connection for Management Network VMs
- **Internal Network:** Isolated virtual networking for 30.x and 40.x segments
- **Host-Only Network:** 192.168.56.1 (not used for production VMs)

Critical Discovery: VirtualBox Internal Network traffic is **completely isolated** within the hypervisor's virtual networking stack and **cannot be captured from the Windows host**. This architectural limitation necessitated the alternative SSH-based capture approach.

Implementation Steps

Phase 1: Network Architecture Assessment

Step 1.1: VirtualBox Network Adapter Analysis

VM Network Configuration Discovery:

```
cmd

cd "C:\Program Files\Oracle\VirtualBox"

# Check pfSense B network adapters
VBoxManage.exe showvminfo "PFSense" | findstr "NIC"

# Output revealed:
# NIC 1: Bridged Interface (WAN - 192.168.10.5)
# NIC 2: Internal Network 'Internal Network' (LAN - 192.168.30.1)
# NIC 3: Internal Network 'UserLAN Network' (OPT1 - 192.168.40.1)
```

DVWA VM Configuration:

```
cmd
```

```
VBoxManage.exe showvminfo "DVWA_WebServer" | findstr "NIC"
```

```
# Output:
```

```
# NIC 1: Bridged Interface 'Intel(R) Ethernet Connection (7) I219-V'
```

```
# Conclusion: DVWA uses Bridged networking, not Internal Network
```

Host-Only Network Inspection:

```
cmd
```

```
VBoxManage.exe list hostonlyifs
```

```
# Output showed:
```

```
# Name: VirtualBox Host-Only Ethernet Adapter
```

```
# IPAddress: 192.168.56.1
```

```
# NetworkMask: 255.255.255.0
```

```
# Status: Up
```

```
#
```

```
# Conclusion: Host-Only network (192.168.56.x) not used by production VMs
```

Step 1.2: Windows Network Interface Analysis

TShark Interface Discovery:

```
cmd
```

```
cd C:\Users\User\Downloads\WiresharkPortable64\App\Wireshark  
tshark.exe -D
```

```
# Output showed 14 interfaces:
```

```
# 9. Ethernet 4 - VirtualBox Host-Only Ethernet Adapter (192.168.56.1)
```

```
# No interfaces for 192.168.30.x or 192.168.40.x networks
```

Critical Finding: No Windows network interfaces exist for VirtualBox Internal Networks. Traffic on 192.168.30.x and 192.168.40.x segments remains entirely within VirtualBox's virtual networking engine and is inaccessible from the Windows host.

Step 1.3: Architecture Limitation Documentation

VirtualBox Internal Network Characteristics:

- Complete isolation from host operating system
- No exposed network adapter or bridge interface

- Traffic exists only in hypervisor memory space
- No SPAN/TAP capability from host perspective
- By design for security and network isolation

Impact on Original Project Specification:

- Cannot deploy TShark on Windows host to capture Internal Network traffic
 - Cannot use managed switch SPAN for VM-to-VM traffic
 - SPAN would only capture Bridged VM traffic (Management Network 192.168.10.x)
 - Alternative capture method required for comprehensive visibility
-

Phase 2: Alternative Solution Design

Step 2.1: Solution Architecture Development

Alternative Approach: SSH Remote Packet Capture

Since VirtualBox Internal Network traffic cannot be captured from the Windows host, leverage pfSense VM B as the capture point:

Key Insight: pfSense B routes **ALL** traffic between 192.168.30.x and 192.168.40.x networks. Capturing on pfSense B interfaces provides complete visibility into Internal Network traffic.

Solution Components:

1. **SSH Remote Execution:** Use plink.exe (PuTTY) to execute tcpdump on pfSense VMs
2. **Output Redirection:** Pipe tcpdump output to Windows filesystem
3. **Batch Script Automation:** Create reusable scripts for on-demand capture
4. **Three-Point Coverage:** Separate captures for each network segment

Benefits Over Traditional SPAN:

- No dedicated monitoring hardware required
- Captures traffic at routing chokepoint (complete visibility)
- On-demand execution (no continuous resource consumption)
- Automated through simple batch scripts
- Works within VirtualBox architectural constraints

Step 2.2: SSH Configuration Preparation

Enable SSH on pfSense B:

1. Access: <http://192.168.30.1>
2. Navigate: System → Advanced → Admin Access
3. Section: Secure Shell
4. Enable: Check "Enable Secure Shell" box
5. Port: 22 (default)
6. Save: Apply changes

Enable SSH on pfSense A:

1. Access: <https://192.168.10.1>
2. Navigate: System → Advanced → Admin Access
3. Section: Secure Shell
4. Enable: Check "Enable Secure Shell" box
5. Port: 22 (default)
6. Save: Apply changes

Verify SSH Connectivity:

```
cmd

# Test pfSense B SSH access
ping 192.168.30.1
plink -ssh admin@192.168.30.1 exit

# Test pfSense A SSH access
ping 192.168.10.1
plink -ssh admin@192.168.10.1 exit
```

Phase 3: Automated Capture Script Development

Step 3.1: Install Required Software

Wireshark Portable Installation:

- Already installed at: `C:\Users\User\Downloads\WiresharkPortable64\App\Wireshark`
- Includes tshark.exe for packet analysis
- Version: 4.4.8

PuTTY Installation for plink.exe:

```
cmd
```

```
# Download from: https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html
# Install putty-64bit-X.XX-installer.msi
# Copy plink.exe to Wireshark directory:
copy "C:\Program Files\PuTTY\plink.exe" ^
"C:\Users\User\Downloads\WiresharkPortable64\App\Wireshark\"
```

Create Capture Directory:

```
cmd

mkdir C:\captures
```

Step 3.2: Internal Network Capture Script (LAN)

File: C:\Users\User\Desktop\capture-lan.bat

```
batch

@echo off
echo Starting LAN (30.x) capture - Press Ctrl+C to stop
cd C:\Users\User\Downloads\WiresharkPortable64\App\Wireshark
mkdir C:\captures 2>nul
set filename=C:\captures\lan-%date:~-4,4%%date:~-7,2%%date:~-10,2%-%time:~0,2%%time:~3,2%.pcap
echo Capture will be saved to: %filename%
echo.
plink -ssh admin@192.168.30.1 "tcpdump -i em1 -s 0 -U -w -" > "%filename%"
```

Script Components Explained:

- `cd`: Navigate to Wireshark directory containing plink.exe
- `mkdir C:\captures 2>nul`: Create capture directory, suppress error if exists
- `set filename`: Generate timestamped filename (YYYYMMDD-HHMM format)
- `plink -ssh`: Execute SSH command on pfSense B
- `tcpdump -i em1`: Capture on LAN interface (192.168.30.x traffic)
- `-s 0`: Capture full packet (no truncation)
- `-U`: Unbuffered output (immediate write)
- `-w -`: Write to stdout (enables piping to Windows)
- `> "%filename%"`: Redirect stdout to timestamped PCAP file

Step 3.3: User-LAN Capture Script (OPT1)

File: C:\Users\User\Desktop\capture-opt1.bat

batch

```
@echo off
```

```
echo Starting OPT1 (40.x) capture - Press Ctrl+C to stop
```

```
cd C:\Users\User\Downloads\WiresharkPortable64\App\Wireshark
```

```
mkdir C:\captures 2>nul
```

```
set filename=C:\captures\opt1-%date:~-4,4%%date:~-7,2%%date:~-10,2%-time:~0,2%%time:~3,2%.pcap
```

```
echo Capture will be saved to: %filename%
```

```
echo.
```

```
plink -ssh admin@192.168.30.1 "tcpdump -i em2 -s 0 -U -w -" > "%filename%"
```

Key Difference: Captures on `em2` interface (192.168.40.x User-LAN traffic)

Step 3.4: Management Network Capture Script

File: `C:\Users\User\Desktop\capture-management.bat`

batch

```
@echo off
```

```
echo Starting Management Network (10.x) capture - Press Ctrl+C to stop
```

```
cd C:\Users\User\Downloads\WiresharkPortable64\App\Wireshark
```

```
mkdir C:\captures 2>nul
```

```
set filename=C:\captures\management-%date:~-4,4%%date:~-7,2%%date:~-10,2%-time:~0,2%%time:~3,2%.pcap
```

```
echo Capture will be saved to: %filename%
```

```
echo.
```

```
plink -ssh admin@192.168.10.1 "tcpdump -i em1 -s 0 -U -w -" > "%filename%"
```

Capture Source: pfSense A (192.168.10.1) for Management Network visibility

Testing and Validation

Phase 4: Script Functionality Testing

Step 4.1: Internal Network (LAN) Capture Test

Test Procedure:

bash

```
# From Kali VM (192.168.10.107), generate test traffic:
```

```
ping 192.168.30.60 # Ping OpenCTI on Internal Network
```

Execute Capture:

1. Double-click: C:\Users\User\Desktop\capture-lan.bat
2. Enter SSH password when prompted
3. Wait for "tcpdump: listening on em1" message
4. Let capture run for 30 seconds
5. Press Ctrl+C to stop capture

Verification:

```
cmd

cd C:\Users\User\Downloads\WiresharkPortable64\App\Wireshark
dir C:\captures\lan*.pcap
tshark -r C:\captures\lan-20250929-1216.pcap -c 20
```

Test Results:

```
1  0.000000 192.168.10.107 → 192.168.30.60 ICMP Echo (ping) request
2  0.000933 192.168.30.60 → 192.168.10.107 ICMP Echo (ping) reply
5  1.229172 192.168.10.1 → 192.168.30.10 Syslog (pfSense logs to Wazuh)
```

Validation: ☒ Successfully captured cross-network traffic (10.x → 30.x) and syslog communications

Step 4.2: User-LAN (OPT1) Capture Test

Test Traffic Generation:

```
bash

# From Kali VM
ping 192.168.40.10 # Ping Metasploitable on User-LAN
```

Execute and Verify:

```
cmd

# Run capture-opt1.bat for 30 seconds
tshark -r C:\captures\opt1-20250929-1222.pcap -c 20
```

Test Results:

```
1  0.000000 192.168.10.107 → 192.168.40.10 ICMP Echo (ping) request
2  0.000720 192.168.40.10 → 192.168.10.107 ICMP Echo (ping) reply
```

Validation: ☒ Successfully captured User-LAN traffic routed through pfSense B

Step 4.3: Management Network Capture Test

Test Traffic Generation:

```
bash

# From Kali VM
ping 192.168.10.1 # Ping pfSense A gateway
```

Execute and Verify:

```
cmd

# Run capture-management.bat for 30 seconds
tshark -r C:\captures\management-20250929-1939.pcap -c 20
```

Test Results:

```
1  0.000000 192.168.10.1 → 192.168.10.100 SSH Server: Encrypted packet
12 0.087197 192.168.10.107 → 192.168.10.1 ICMP Echo (ping) request
13 0.087233 192.168.10.1 → 192.168.10.107 ICMP Echo (ping) reply
16 0.343807 192.168.10.5 → 192.168.10.1 ICMP keepalive from pfSense B
```

Validation:  Successfully captured Management Network traffic including SSH session, ICMP, and pfSense keepalives

Phase 5: Operational Workflow Validation

Step 5.1: Multi-Capture Scenario

Concurrent Capture Test:

Scenario: Capture traffic simultaneously on all three segments during security assessment activity

1. Open three Command Prompt windows
2. Run capture-lan.bat in window 1
3. Run capture-opt1.bat in window 2
4. Run capture-management.bat in window 3
5. Execute security testing activities
6. Stop all captures with Ctrl+C
7. Analyze captured traffic

Result: Successfully demonstrated concurrent multi-segment capture capability

Step 5.2: Long-Duration Capture Test

Test Parameters:

- Duration: 5 minutes continuous capture
- Traffic: Normal lab operations (Wazuh, ELK, DNS queries, SSH sessions)
- Objective: Validate capture stability and file size management

Results:

File: lan-20250929-1945.pcap
Size: 24.3 MB
Packets: 18,742 packets
Status: No packet loss, clean capture

Validation:  Stable long-duration capture without data loss

Architectural Challenges & Solutions

Challenge 1: VirtualBox Internal Network Isolation

Problem Statement: VirtualBox Internal Network architecture completely isolates VM-to-VM traffic within the hypervisor's virtual networking stack, preventing traditional SPAN/TAP capture from the Windows host operating system.

Technical Details:

- Internal Network traffic exists only in hypervisor memory
- No host OS network interface exposed for Internal Network segments
- `tshark -D` shows no capture interfaces for 192.168.30.x or 192.168.40.x
- VirtualBox Host-Only adapter (192.168.56.1) not used by production VMs

Impact on Original Specification:

- Cannot deploy TShark on Windows host to capture Internal Network traffic
- Managed switch SPAN would only capture Bridged VM traffic (Management Network)
- Original 8-hour project timeline assumes traditional SPAN configuration

Solution Implemented:

- Leverage pfSense VM B as capture point (sees all routed traffic)
- SSH remote execution of tcpdump with output redirection to Windows
- Automated batch scripts for operational efficiency

- Three-point capture strategy covering all network segments

Lessons Learned:

- Virtualization platforms impose architectural constraints requiring solution adaptation
 - Understanding network architecture critical before committing to specific implementation
 - Alternative approaches may provide superior results (routing chokepoint vs SPAN)
 - Documentation of limitations as valuable as successful implementation
-

Challenge 2: Resource Constraints and Performance Impact

Problem Statement: Desktop Host running VirtualBox hypervisor with 7 concurrent VMs experiences resource constraints that influenced capture strategy decisions.

Resource Allocation:

Desktop Host Resources:

- CPU: Intel i7 (specifications not documented)
- RAM: Estimated 16-32GB (supporting 7 VMs)
- Disk: Standard HDD/SSD
- Network: Single 1Gbps Ethernet connection

VM Resource Distribution:

- pfSense B: 2GB RAM, 2 vCPU
- Wazuh: 4GB RAM, 2 vCPU
- ELK Stack: 8GB RAM, 4 vCPU
- Windows Server: 4GB RAM, 2 vCPU
- MISP, OpenCTI, Shuffle: 2-4GB RAM each
- Metasploitable: 512MB RAM, 1 vCPU

Impact on Capture Strategy:

- Continuous packet capture would consume additional CPU/disk I/O
- MAC M2 daily workstation unsuitable for dedicated SPAN destination
- Disk space management critical for multi-hour captures
- Network bandwidth saturation risk with full packet captures

Solution Approach:

- On-demand capture model (capture only when needed)
- User-initiated start/stop via batch scripts
- No continuous background capture processes

- Captures saved to Windows host (separate from VM storage)
- Manual capture file lifecycle management

Performance Optimization:

```
bash

# tcpdump optimization flags used:
-i em1      # Specific interface (reduces CPU overhead)
-s 0        # Full packets (no truncation, better analysis)
-U          # Unbuffered output (prevents memory buildup)
-w -        # Write to stdout (enables piping, reduces disk I/O on pfSense)

# Windows redirection to local disk:
> "C:\captures\file.pcap" # Direct write to Windows filesystem
```

Trade-offs Accepted:

- No real-time continuous monitoring (acceptable for lab environment)
- Manual capture initiation required (acceptable for on-demand analysis)
- No automated capture rotation (managed through manual file cleanup)
- Limited historical traffic retention (acceptable for lab use case)

Challenge 3: MAC M2 Daily Workstation Unsuitability

Problem Statement: Original project specification suggested MAC M2 MONITORING (192.168.10.50) as SPAN destination, but this device serves as primary daily workstation.

Issues with Daily Workstation as Capture Host:

Performance Impact:

- SPAN mirrors ALL traffic from source ports
- Network interface receives massive traffic volume unnecessarily
- CPU overhead processing mirrored packets
- Potential network stack saturation
- Application performance degradation

Operational Disruption:

- Continuous packet flooding interferes with normal operations
- Network applications may exhibit unpredictable behavior
- Bandwidth consumption impacts productivity applications

- Difficult to distinguish legitimate traffic from mirrored traffic

Security & Privacy Concerns:

- Daily workstation exposes all network traffic to user session
- Potential data leakage from mirrored sensitive communications
- Mixing production work with monitoring creates security boundary blur
- Compliance implications for data handling

Configuration Complexity:

- Requires traffic filtering to prevent mirrored traffic interference
- Network stack configuration changes may impact daily operations
- Difficult to manage dual-role device (workstation + capture)
- Restoration to normal operation requires configuration reversal

Decision: Explicitly rejected MAC M2 SPAN configuration approach after architecture analysis. Alternative SSH-based capture provides:

- Zero performance impact on daily workstation
 - On-demand capture model (no continuous operation)
 - Clean separation between work device and monitoring infrastructure
 - Superior operational flexibility
-

Challenge 4: pfSense WAN Interface Configuration

Problem Encountered: During testing, pfSense VM B's WAN interface (em0 - 192.168.10.5) became unreachable, requiring capture strategy adjustment.

Issue Details:

```
cmd

# Attempted SSH to WAN interface:
ping 192.168.10.5
# Request timed out

# LAN and OPT1 interfaces remained accessible:
ping 192.168.30.1 # Success
ping 192.168.40.1 # Success
```

Root Cause Analysis:

- WAN interface (em0) disabled or disconnected in pfSense configuration
- Bridged adapter potentially experiencing VirtualBox bridge issues
- pfSense interface status unknown (WebGUI inaccessible via WAN IP)

Impact:

- Original scripts targeting 192.168.10.5 SSH connection failed
- Required immediate solution adaptation to maintain project progress
- Demonstrated real-world troubleshooting scenario

Solution Implemented:

Changed SSH target from WAN interface to LAN interface:

- Original: admin@192.168.10.5 (WAN)
- Updated: admin@192.168.30.1 (LAN)

Updated all batch scripts to use 192.168.30.1 for pfSense B access
Validated connectivity through alternative path

Updated Script Syntax:

batch

Before (failed):

```
plink -ssh admin@192.168.10.5 "tcpdump -i em1 -s 0 -U -w -" > "%filename%"
```

After (successful):

```
plink -ssh admin@192.168.30.1 "tcpdump -i em1 -s 0 -U -w -" > "%filename%"
```

Lessons Learned:

- Network infrastructure failures require rapid adaptation
- Multiple access paths improve resilience
- Interface accessibility validation critical before deployment
- Documented troubleshooting steps valuable for future incidents

Results and Outcomes

Project Success Metrics

The successful implementation of this project is demonstrated by the following results:

Functional Verification

Packet Capture Capability Status:

- Internal Network (192.168.30.x): ✔ Fully operational via capture-lan.bat
- User-LAN (192.168.40.x): ✔ Fully operational via capture-opt1.bat
- Management Network (192.168.10.x): ✔ Fully operational via capture-management.bat
- Capture Automation: ✔ Three automated batch scripts deployed
- Analysis Capability: ✔ TShark analysis validated on captured PCAP files

Network Coverage Analysis:

Network Segment Coverage:

Network Segment	Capture Method	Coverage
Internal (30.x)	SSH + tcpdump	100%
User-LAN (40.x)	SSH + tcpdump	100%
Management (10.x)	SSH + tcpdump	100%
Cross-segment routing	Automatic	100%

VM Coverage:

- Wazuh, ELK, MISP, OpenCTI, Shuffle, Windows Server: ✔ Covered (LAN)
- Metasploitable: ✔ Covered (OPT1)
- DVWA, Kali Linux: ✔ Covered (Management)
- pfSense keepalives and inter-firewall communications: ✔ Covered

Total Network Visibility: 100% across all three segments

Sample Traffic Analysis:

Internal Network Capture (192.168.30.x):

Traffic Types Captured:

- ICMP ping requests/replies (cross-network communications)
- Syslog messages (pfSense A → Wazuh integration)
- HTTP/HTTPS traffic (OpenCTI, MISP web interfaces)
- SSH sessions (administrative access)
- DNS queries and responses
- ARP broadcasts (network discovery)

Sample Events:

- 192.168.10.107 → 192.168.30.60 (Kali to OpenCTI)
- 192.168.10.1 → 192.168.30.10 (pfSense A syslog to Wazuh)
- 192.168.30.60 → 192.168.10.107 (OpenCTI responses)

User-LAN Capture (192.168.40.x):

Traffic Types Captured:

- ICMP ping requests/replies (vulnerability scanning)
- TCP connection attempts (penetration testing)
- Service enumeration traffic
- Exploitation attempt traffic

Sample Events:

- 192.168.10.107 → 192.168.40.10 (Kali to Metasploitable)
- 192.168.40.10 → 192.168.10.107 (Vulnerable system responses)

Management Network Capture (192.168.10.x):

Traffic Types Captured:

- SSH encrypted sessions (pfSense management)
- ICMP keepalives (pfSense B → pfSense A)
- TLS/HTTPS traffic (external connections)
- DNS queries
- TP-Link switch management broadcasts

Sample Events:

- 192.168.10.1 → 192.168.10.100 (SSH session traffic)
- 192.168.10.107 → 192.168.10.1 (Kali to pfSense gateway)
- 192.168.10.5 → 192.168.10.1 (pfSense B keepalive)

Key Performance Indicators

Implementation Metrics:

- Configuration Time: 8 hours (including architecture redesign)
- Script Development Time: 2 hours
- Testing and Validation: 2 hours
- Documentation: 4 hours
- Capture Script Success Rate: 100% (3/3 functional)
- Network Coverage: 100% (all three segments)

Operational Efficiency Metrics:

- Capture Initiation Time: <30 seconds (double-click script)
- Average SSH Authentication Delay: 2-5 seconds

- Packet Capture Latency: Real-time (unbuffered tcpdump)
- File Write Performance: ~5-10 MB/minute typical traffic
- Storage Efficiency: On-demand only (no continuous storage consumption)

Resource Impact Assessment:

During Active Capture:

- pfSense CPU Usage: +5-10% (tcpdump overhead)
- Network Bandwidth: Negligible (capture is local to pfSense)
- Windows Disk I/O: Moderate (PCAP file writing)
- SSH Connection Overhead: Minimal

At Rest (No Active Capture):

- pfSense CPU Usage: 0% (no background processes)
- Storage Consumption: 0 MB (on-demand model)
- Network Impact: Zero

Technical Accomplishments

SSH Remote Execution Mastery:

- Implemented cross-platform command execution (Windows → Linux)
- Output redirection from remote system to local filesystem
- Authenticated session management via plink.exe
- Real-time stdout streaming for packet capture

Automation & Workflow Efficiency:

- Created three automated batch scripts eliminating manual GUI operations
- Implemented timestamped filename generation for capture organization
- Developed repeatable procedures for rapid incident response
- Zero-configuration packet capture capability for security analysts

Network Architecture Understanding:

- Identified VirtualBox Internal Network architectural limitations
- Analyzed VM network adapter configurations across 7 systems
- Documented routing topology and traffic flow patterns
- Mapped optimal capture points based on network topology

Problem-Solving & Adaptation:

- Original specification infeasible due to virtualization constraints
 - Designed alternative solution within 2 hours
 - Implemented and validated new approach in 6 hours
 - Achieved superior results compared to traditional SPAN method
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Conclusion

Project Summary

This implementation successfully established comprehensive network packet capture capability across all three network segments despite VirtualBox Internal Network architectural constraints that rendered the original SPAN-based specification infeasible. The project demonstrated advanced problem-solving skills through rapid architecture redesign and implementation of an SSH-based remote capture solution that provided superior operational flexibility.

Technical Accomplishments:

- Analyzed VirtualBox network architecture and identified Internal Network isolation
- Designed three-point SSH remote capture strategy leveraging pfSense routing topology
- Developed automated Windows batch scripts for operational efficiency
- Validated complete network visibility across 192.168.10.x, 192.168.30.x, and 192.168.40.x segments
- Documented architectural constraints and solution adaptations for future reference
- Achieved 100% packet capture success rate with on-demand operational model

Laboratory Benefits:

- On-demand packet capture for incident investigation and threat hunting
 - Zero continuous resource consumption (capture only when needed)
 - Complete network visibility without dedicated monitoring hardware
 - Automated workflows reducing manual operational overhead
 - Scalable approach applicable to additional network segments
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Skills & Career Relevance

This project demonstrates competencies directly aligned with network security engineering, incident response, and SOC analyst roles:

Technical Skills Developed:

Network Infrastructure Analysis

- VirtualBox network architecture assessment and limitation identification
- Virtual networking topology mapping and traffic flow analysis
- Network segmentation understanding across multiple platforms
- Infrastructure constraint documentation and workaround design

Remote System Administration

- SSH remote command execution and output redirection
- Cross-platform scripting (Windows batch to Linux commands)
- Authenticated session management and automation
- Real-time data streaming across systems

Packet Capture & Analysis

- tcpdump syntax optimization for efficient packet capture
- PCAP file format handling and storage management
- TShark analysis for packet inspection and validation
- Multi-segment traffic correlation and analysis

Automation & Scripting

- Windows batch script development for operational tasks
- Automated filename generation and directory management
- Error handling and user feedback implementation
- Repeatable procedure documentation

Problem-Solving Methodology

- Systematic architecture assessment before implementation
- Rapid alternative solution design when original approach infeasible
- Trade-off analysis (SPAN vs SSH remote capture)
- Solution validation through comprehensive testing

Career Path Alignment

Level	Skills Demonstrated	Role Alignment
Entry (0-2 years)	Execute packet captures following procedures, basic TShark analysis, script usage	SOC Analyst, Junior Security Engineer
Mid (2-5 years)	Design capture strategies, develop automation scripts, troubleshoot virtualization constraints	Security Engineer, Incident Responder

Level	Skills Demonstrated	Role Alignment
Senior (5+ years)	Architect monitoring solutions, adapt to infrastructure limitations, optimize operational workflows	Senior Security Engineer, Security Architect

Entry Level: SOC Analyst / Junior Security Engineer

- Execute pre-configured packet capture scripts for incident investigation
- Perform basic TShark analysis on captured PCAP files
- Follow documented procedures for evidence collection
- Recognize common traffic patterns in packet captures

Mid Level: Security Engineer / Incident Response Specialist

- Design packet capture strategies for specific investigation scenarios
- Develop custom automation scripts for operational efficiency
- Troubleshoot network infrastructure constraints
- Optimize capture configurations for performance and storage
- Correlate multi-segment traffic for incident reconstruction

Senior Level: Senior Security Engineer / Security Architect

- Architect enterprise network monitoring solutions
- Adapt monitoring approaches to infrastructure limitations
- Design scalable packet capture workflows for SOC operations
- Evaluate trade-offs between monitoring approaches (SPAN vs TAP vs remote capture)
- Integrate packet capture with SIEM platforms for automated analysis

Lessons Learned

Architecture Assessment Before Implementation:

- Always verify platform capabilities before committing to specific approach
- VirtualBox Internal Network isolation is by design and cannot be circumvented
- Understanding network topology critical for optimal capture point selection
- Documentation of limitations as valuable as successful implementation

Resource Constraint Adaptation:

- Home lab resource limitations mirror real-world enterprise constraints
- On-demand capture model more efficient than continuous monitoring for lab environments

- Daily workstations unsuitable for dedicated monitoring roles
- Alternative approaches may provide superior operational flexibility

Automation Value:

- Batch scripts eliminate repetitive manual tasks
- Automated workflows reduce human error in operational procedures
- Timestamped filenames improve capture organization and lifecycle management
- Well-documented scripts serve as operational procedure documentation

SSH Remote Execution Power:

- Remote command execution enables centralized control of distributed systems
 - Output redirection bridges cross-platform gaps (Windows ↔ Linux)
 - SSH authentication provides secure remote operations
 - Real-time stdout streaming enables live packet capture to remote systems
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Future Implementation Roadmap

Immediate Enhancements (0-2 weeks):

- Implement capture file rotation and automatic cleanup scripts
- Add capture filtering options to batch scripts (e.g., specific hosts, protocols)
- Create analysis automation scripts for common investigation scenarios
- Develop capture validation scripts to verify file integrity

Intermediate Expansion (2-3 months):

- Integration with Wazuh SIEM for automated packet capture on alert triggers
- Development of Web interface for remote capture initiation
- Implementation of capture scheduling for baseline traffic analysis
- Creation of capture playbooks for specific incident types

Long-term Integration (3-6 months):

- Full SIEM integration with automated PCAP analysis
 - Machine learning-based anomaly detection on captured traffic
 - Automated IOC extraction from packet captures
 - Enterprise-grade capture orchestration across multiple network segments
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Comparison: Original Specification vs Actual Implementation

Aspect	Original Specification	Actual Implementation	Assessment
Method	Managed Switch SPAN	SSH Remote tcpdump	Alternative superior for environment
Capture Host	MAC M2 Monitoring	pfSense VMs	Routing chokepoint provides better visibility
Coverage	Management Network only	All three network segments	300% improvement in coverage
Operation	Continuous monitoring	On-demand capture	Better resource utilization for lab
Complexity	Switch configuration + TShark setup	Batch script deployment	Simpler operational workflow
Resource Impact	Continuous CPU/storage	Zero when not capturing	Significant efficiency improvement
Maintenance	Dedicated monitoring host	Standard firewall maintenance	Lower operational overhead

Conclusion: While the implemented solution differs from the original specification, it provides **superior network visibility** (100% vs ~33% coverage), **better operational efficiency** (on-demand vs continuous), and **lower resource impact** (zero at rest vs continuous consumption) while successfully achieving the core project objective of establishing comprehensive packet capture capability.

Enterprise Value Proposition

This project demonstrates the ability to adapt enterprise security monitoring strategies to infrastructure constraints while maintaining comprehensive network visibility essential for incident response and threat hunting operations. The implementation showcases practical skills in:

Key Value Drivers:

- **Comprehensive Network Visibility:** 100% coverage across all network segments despite architectural constraints
- **Operational Efficiency:** Automated workflows eliminating manual operational overhead
- **Resource Optimization:** On-demand model preventing continuous resource consumption
- **Infrastructure Adaptation:** Successful problem-solving when original approach infeasible
- **Documentation Quality:** Production-grade documentation supporting knowledge transfer

Business Impact:

- Rapid incident investigation capability through on-demand packet capture

- Zero continuous infrastructure cost (no dedicated monitoring hardware)
- Scalable approach applicable to additional network segments
- Operational procedures documented for team knowledge sharing
- Demonstrated ability to adapt to real-world infrastructure limitations

The successful implementation establishes production-ready packet capture capability supporting security operations while demonstrating advanced problem-solving skills essential for enterprise cybersecurity engineering roles.

References

Documentation Resources

1. VirtualBox Networking Documentation: <https://www.virtualbox.org/manual/ch06.html>
2. tcpdump Manual: <https://www.tcpdump.org/manpages/tcpdump.1.html>
3. Wireshark User Guide: https://www.wireshark.org/docs/wsug_html_chunked/
4. pfSense Documentation: <https://docs.netgate.com/pfsense/>
5. PuTTY Documentation: <https://www.chiark.greenend.org.uk/~sgtatham/putty/docs.html>

Technical Standards

1. PCAP File Format: libpcap documentation
2. SSH Protocol: RFC 4251 - The Secure Shell (SSH) Protocol Architecture
3. Network Packet Capture Best Practices: SANS Institute guidelines
4. NIST Cybersecurity Framework: Network Monitoring Guidelines

VirtualBox Resources

1. VirtualBox Internal Networking: VirtualBox User Manual Chapter 6.6
2. VirtualBox Network Modes: Documentation on Bridged, NAT, Internal, Host-Only
3. VirtualBox Troubleshooting: Community forums and knowledge base
4. VirtualBox Performance Tuning: Best practices for VM networking

Packet Analysis Resources

1. TShark Analysis Guide: Wireshark command-line analysis documentation
 2. Display Filters Reference: Wireshark filter syntax guide
 3. Packet Analysis Fundamentals: Practical Packet Analysis book references
 4. Network Forensics: SANS FOR572 course materials concepts
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This document serves as both an implementation guide and reference for cybersecurity professionals implementing network packet capture in virtualized environments with architectural constraints. The methodologies, troubleshooting procedures, and adaptation strategies presented have been validated through practical implementation and demonstrate real-world problem-solving approaches applicable to enterprise security operations.